Learning-Based Adaptive Control of a Hydraulic Manipulator

Fully Rebuilt Control Architecture, Learned Models, PPO Formulation, Experimental Plan, and Thesis Outline

Current actuator baseline:	**PPO–ROM**
Non-adaptive learned baseline:	**PPO–ID–FD**
Proposed online-adaptive method:	**PPO–ID(AAM)–FD**

Current implementation status. The privileged-latent evidence chain through Stage 22 is complete and the strict latent-matters gate has passed. The online Arm Adaptation Module (AAM), the latent-conditioned inverse-model adapter, and the final three-policy comparison are planned next; they must not be described as completed results.

Prepared as the working implementation and evaluation plan

Full rebuild v2: 9 August 2026

Contents

1 Purpose of this revision **3**
1.1 Authoritative final experimental matrix . . . 3
1.2 What is already established and what remains open . . . 4

2 System and plant definition **4**
2.1 Physical system represented in simulation . . . 4
2.2 Primary control task . . . 4
2.3 Multi-rate actuator interface . . . 5
2.4 Deployment-domain factors . . . 5

3 Current learned-model inventory **5**
3.1 Correct relationship among ROM, SD, encoder and AAM . . . 6

4 Data and Stage-22 evidence chain **7**
4.1 Two data provenances . . . 7
4.2 Final frozen dataset . . . 7
4.3 Meaning of the strict Stage-22 pass . . . 7

5 The three final control stacks **7**
5.1 PPO–ROM: certified actuator baseline . . . 7
5.2 PPO–ID–FD: non-adaptive learned baseline . . . 7
5.3 PPO–ID(AAM)–FD: proposed adaptive method . . . 8

6 Current PPO formulation **8**
6.1 Policy role and action contract . . . 8
6.2 Observation contract . . . 8
6.3 Reward and acceptance are separate . . . 9
6.4 Clipped PPO objective . . . 9
6.5 Training protocol and the distillation finding . . . 9

7 Research questions and testable hypotheses **10**

8 Remaining implementation gates **10**

9 Final comparative experiments **11**
9.1 Fixed comparison matrix . . . 11
9.2 System response . . . 12
9.3 Delay . . . 12
9.4 Nonlinearity . . . 12
9.5 End-effector and mowing performance . . . 12

9.6 Robustness and generalization 12
9.7 Adaptation ablations 12
9.8 Statistics 13

10 Updated thesis chapter plan **13**

11 Claim discipline and reproducibility checklist **14**

12 Current sources of truth **14**

1 Purpose of this revision

This plan is rebuilt from the maintained repository contracts and the completed adaptation evidence. It supersedes every earlier experimental matrix and the former assumption that the AAM would condition a reduced-order model directly.

The maintained comparison uses three descriptive stacks:

1. **PPO–ROM**: the policy commands desired cylinder motion, which is executed by the validated command-conditioned hydraulic ROM and PhysX;
2. **PPO–ID–FD**: the same policy interface drives a frozen inverse dynamics (ID) model followed by the validated strict forward dynamics (FD) model and PhysX;
3. **PPO–ID(AAM)–FD**: the AAM infers a dynamics latent from causal observable history, and that latent conditions a small residual adapter around the frozen ID model before the same frozen FD and PhysX.

The scientific variable in the final comparison is therefore the actuator and adaptation path. The task, policy observation and action contracts, terrain, reward, training budget, test factors, seeds, and acceptance gates must otherwise remain matched.

1.1 Authoritative final experimental matrix

The following three rows are the complete final matrix. There are no additional policy variants in the primary thesis comparison.

Stack	Actuation path	Online adaptation	Current status
PPO–ROM	policy → desired cylinder motion → command ROM → PhysX	None	Certified baseline; passes all eight gates on six held-out meshes.
PPO–ID–FD	policy → desired motion → frozen ID → frozen strict FD → PhysX	None	Implemented baseline; better lateral tracking but fails the worst-mesh contact gate.
PPO–ID(AAM)–FD	policy → desired motion → frozen ID + adapter($\hat{z}$) → frozen strict FD → PhysX	Causal AAM history → $\hat{z}$	Proposed method; Stages 23–31 and final control evaluation remain to be completed.

1.2 What is already established and what remains open

Status	Established evidence	Interpretation
Completed	Command ROM, control ID, strict FD, IsaacLab/PhysX plant interface, matched randomization design, 118-run frozen dataset, no-latent system-dynamics model, factor encoder and latent-conditioned system-dynamics model.	These are available components or completed identification results.
Passed	Stage 22: on the reserved factor configurations, true latent content improves extension and velocity rollout over zero, shuffled, random, nominal and protocol-matched no-latent controls.	The privileged latent contains useful dynamics information. This is a prerequisite for online adaptation, not yet proof of control improvement.
Next	Freeze the Stage-22 representation, build causal AAM histories, train the AAM, validate downstream rollout retention, test ID transfer, then train the small latent-conditioned ID adapter.	Only after these gates pass may the deployable adaptive stack be compared with the two frozen baselines.

2 System and plant definition

2.1 Physical system represented in simulation

The system is the MULAG FME 500 truck-mounted hydraulic manipulator and mower head. Four hydraulic cylinder channels position the arm and head. IsaacLab/PhysX supplies the multibody mechanics, contacts, joint limits, payload effects, chassis motion, terrain, and cutter–ground interaction. Learned hydraulic models supply the actuator transition that converts the policy's cylinder-motion intent into the motion applied to PhysX.

The thesis must distinguish three objects that were previously blurred together:

- the **mechanical plant**: arm, truck, cutter, terrain and contact dynamics in PhysX;
- the **hydraulic actuator representation**: command ROM or ID–FD;
- the **system-dynamics surrogate**: the Stage-19/21 SD models used to test whether privileged factor information predicts future hydraulic state. The SD is not the final actuator used by a PPO stack.

2.2 Primary control task

The final comparison is the maintained verge-following task. The truck follows the road at 4 km/h while the four-cylinder arm keeps the cutter on the verge centre line, near the target cut height, in ground contact, head-down, and within mechanical limits. Terrain is a disturbance that makes tracking nontrivial; it is not an unconstrained obstacle course. All eleven reference-path families are restricted to the measured actuator envelope. At the maintained settings the optimistic lateral

cylinder-rate ceiling is 0.200 m/s, and preflight testing rejects path families that demand more authority.

The parked line, arc, sine, circle, and square workflow remains useful for plant and actuator-model qualification. It is not a fourth policy stack and should not be mixed with the final moving-verge statistics unless explicitly reported as a separate diagnostic experiment.

2.3 Multi-rate actuator interface

The policy acts through the environment's four-dimensional cylinder command port. The verge task uses a 20 ms policy/control step. The command ROM has a native 50 ms learned transition; its runtime predicts at the model grid and distributes the resulting motion over the faster PhysX updates. Stage-17 learning samples use exact 50 ms transitions at control indices $0, 3, 6, \ldots$ and never interpolate. The thesis must report physics, policy, learned-model, AAM-update and logging rates separately rather than calling all of them the "control frequency."

For the command-ROM plant, the external input is desired cylinder extension change and velocity. Valve actuation is inferred internally by the ROM. For ID–FD, the ID explicitly maps desired motion and state to valve command, and the strict FD maps valve/state to cylinder extension increment. This difference is the actuator variable being tested.

2.4 Deployment-domain factors

The privileged physical-factor vector is

$$e = [e_{\text{flow}}, e_{\text{compliance}}, e_{\text{seal}}, e_{\text{pressure}}, e_{\text{mass}}, e_{\text{COM}z}]. \tag{1}$$

The final ranges are respectively $[-0.25, 0.25]$, $[-0.25, 0.10]$, $[0, 1]$, $[-0.15, 0.15]$, $[0.8, 1.5]$, and $[-0.10, 0.05]$. Hydraulic-only, mechanical-only, combined-training and unseen combined-test domains are kept distinct. One factor vector is fixed for a complete identification run. Emergent delay is measured as a response property; it is not supplied as a known randomized factor.

3 Current learned-model inventory

Component	Contract	Current role	Important qualification
Command ROM	desired cylinder motion and current hydraulic state → internal valve response and cylinder transition	Frozen actuator in **PPO–ROM**; also the deployment plant used to collect the new AAM/SD dataset	Runs on the 50 ms model grid. It is not externally valve-commanded and is not the model that the AAM will condition.
Control ID	desired cylinder target/motion and state → valve command	Frozen inverse model in both ID–FD stacks	The adaptive method adds a bounded residual adapter around this frozen model; it does not retrain the base ID initially.

Component	Contract	Current role	Important qualification
Strict FD	valve command and state/history → cylinder extension increment	Frozen forward model in both ID–FD stacks	Use the rank-promoted strict checkpoint. The older control FD underused valve information and is not the validated comparison model.
SD_0	observable history/state/action → next 12-D hydraulic-state delta	Matched no-latent identification baseline	Uses hidden pressure input, history-six belief, joint context and ten-step unrolling. Reserved-test $R^2_{all} = 0.7989$.
$\mu(e)$ and SD_z	factor vector $e \to z$; state/action/history/$z \to$ next state delta	Privileged representation and latent-matters test	Selected latent dimension $d_z = 2$. Reserved-test $R^2_{all} = 0.8186$. Frozen after Stage 23 and never used as a deployable sensor.
AAM	causal observable history $H_t \to \hat{z}_t$	Planned deployment-time context estimator	Not yet trained. It may use applied action, cylinder extension/velocity and joint state/history, but not e, true z, future samples, hidden ROM state or hidden differential pressure.
ID adapter	$(s_t, u_{\mathrm{ID}}, \hat{z}_t) \to \Delta u_t$	Planned adaptation mechanism in **PPO–ID(AAM)–FD**	Small, bounded and residual. Train first with true z, then replace by $\hat{z}$; preserve a safe fallback to frozen ID.

3.1 Correct relationship among ROM, SD, encoder and AAM

The command ROM is an existing actuator representation learned from historical Simscape data. The new SD_0, SD_z and encoder are trained from scratch on randomized IsaacLab runs generated through the command-ROM-plus-PhysX deployment plant. Their purpose is to define and validate a compact privileged dynamics representation. The AAM then learns to infer that representation from deployable history. In the first online-control design, the inferred latent changes the ID command through a small adapter; it does not modify the command ROM or the FD. Direct policy conditioning on $\hat{z}$ is a later ablation.

This separation prevents three invalid claims:

- Stage 22 does not show that the AAM works, because Stage 22 uses true privileged z rather than $\hat{z}$;
- a model trained on randomized data is robust, not automatically adaptive;
- a latent dimension must not be named after a physical parameter unless the relationship is measured rather than inferred visually.

4 Data and Stage-22 evidence chain

4.1 Two data provenances

Historical Simscape data produced the frozen command ROM and the frozen ID/FD artifacts. Those artifacts are retained as comparison components. The current adaptation experiment does not collect new Simscape data and does not reuse an old AAM or system-dynamics model.

The new adaptation dataset was collected in IsaacLab using the command ROM plus PhysX, the same four-dimensional command port used at deployment, DLS task motion, and dedicated independent four-channel PRBS windows. DLS closes the loop on measured cylinder state; no joint-target bypass or PPO identification policy is used.

4.2 Final frozen dataset

The final collection contains 118 complete runs: 2 nominal, 40 hydraulic-only, 12 mechanical-only, 48 combined-training and 16 unseen combined-test. Four vectorized Isaac environments were used with per-environment factors and logging. Unsafe or invalid slots were discarded and retried individually. The design has 116 unique factor keys (98.3%), full active-factor rank, populated quartiles and disjoint factor keys across train, validation and test.

The frozen split is 82 training, 20 validation and 16 reserved test runs. Windows never cross runs. The hydraulic state is

$$s_t = [x_t^{1:4}, \dot{x}_t^{1:4}, \Delta p_t^{1:4}] \in \mathbb{R}^{12}, \tag{2}$$

with joint position and velocity used only as context. The prediction target is the next 50 ms state delta. Differential pressure is recorded as a target and diagnostic but hidden from the final deployment-mirror SD input, making the causal history belief necessary.

4.3 Meaning of the strict Stage-22 pass

The wide factor ranges and matched excitation corrected the weak-identifiability problem in earlier datasets. At a 20-step rollout horizon, paired bootstrap confidence intervals show that true z improves both extension and velocity over zero, shuffled, random, nominal and protocol-matched no-latent controls, pooled and on the 16 reserved test runs. This justifies freezing the two-dimensional representation and attempting online inference. It does not permit skipping the AAM, ID-transfer or closed-loop control gates.

5 The three final control stacks

5.1 PPO–ROM: certified actuator baseline

$$\text{policy} \rightarrow \text{desired cylinder motion} \rightarrow \text{command ROM} \rightarrow \text{PhysX}$$

This is the current shippable verge stack. On six held-out terrain meshes it passes all eight task and safety gates. Its worst-mesh worst-family lateral error is 0.0817 m and its worst ground-contact fraction is 0.7854. The command ROM is frozen for the final comparison.

5.2 PPO–ID–FD: non-adaptive learned baseline

$$\text{policy} \rightarrow \text{desired cylinder motion} \rightarrow \text{frozen ID} \rightarrow \text{frozen strict FD} \rightarrow \text{PhysX}$$

The environment subclasses the ROM task and changes only the actuator path. On the same six meshes, ID–FD improves worst lateral tracking to 0.0628 m (23% better) but reaches only 0.6882 ground contact on the worst mesh and therefore fails the 0.700 contact gate. This is a useful negative result: the ID–FD model gives finer lateral response but poorer vertical ground following. It is a strong baseline, not a certified deployment result.

5.3 PPO–ID(AAM)–FD: proposed adaptive method

$$\begin{gathered}\text{observable history} \rightarrow \text{AAM} \rightarrow \hat{z}_t \\ \text{policy intent} \rightarrow \text{frozen ID} + \text{adapter}(\hat{z}_t) \rightarrow \text{frozen strict FD} \rightarrow \text{PhysX}\end{gathered}$$

The first proposed contribution is deliberately small and testable. The frozen ID produces u_{base} and the adapter produces a bounded correction:

$$u_{\text{base},t} = ID_{\text{frozen}}(s_t, r_t), \tag{3}$$

$$u_t = \text{clip}(u_{\text{base},t} + g_\psi(s_t, r_t, u_{\text{base},t}, \hat{z}_t)), \tag{4}$$

$$\hat{z}_t = AAM_\omega(H_t). \tag{5}$$

The FD, PhysX plant and base ID remain frozen initially. During warm-up or invalid AAM output, the correction is zero and the stack falls back to frozen ID–FD.

6 Current PPO formulation

6.1 Policy role and action contract

PPO is an upper-level residual cylinder-motion controller. It does not command wheels, direct joint torques or an unbounded valve signal. The actor emits four actions, one for each hydraulic cylinder. The same normalized action contract is used for every stack. At the maintained final terrain stage, cylinders 1–3 have 0.004 m command scale per unit action and 0.018 m target-lead caps; the head-pitch cylinder has 0.012 m scale and a 0.30 m lead cap. Limit shielding, rate handling and clipping keep the residual within the learned actuator envelope.

The nominal DLS controller remains the privileged teacher and identification controller. It provides structured task-to-cylinder behavior and safe excitation trajectories. It is not the online adaptation mechanism. The deployable actor is a memoryless Gaussian MLP; temporal adaptation belongs to the AAM history estimator, not to an unreported privileged policy input.

6.2 Observation contract

The matched final policy observation is 66-dimensional. Its groups are:

- nine forward terrain-height profile samples;
- cut-height and lateral errors and cutter vertical velocity;
- four cylinder extensions, four velocities and low/high headroom;
- previous applied action, current inferred valve command, actuator-predicted extension increment and commanded target lead;
- head alignment and alignment rate, truck speed, scan validity, signed head-axis components, cutter contact force and a three-point path preview;

- twelve model-error/load channels: cylinder tracking residual, previous actuator prediction residual and normalized cylinder effort.

No trajectory-family label or privileged factor vector enters the actor. If the optional AAM-to-policy ablation is performed, its added latent input must be declared explicitly and cannot be compared as though the policy architecture were unchanged.

6.3 Reward and acceptance are separate

The maintained reward has nine bounded terms with live gradients beyond the acceptance boundary. Positive task terms are cut height, lateral tracking and ground contact. Safety and regularization terms penalize head misalignment, contact loss, action rate, tool acceleration, contact force and loss of cylinder headroom. The published weight budget is $+22$ for task terms and -20 maximum magnitude for safety terms. Action magnitude is excluded because sustained offsets are required by several path families.

The reward is not the result criterion. Promotion uses eight physical gates evaluated at the worst of at least three distinct terrain meshes: cut-height error ≤ 0.040 m, lateral error ≤ 0.300 m, ground contact ≥ 0.700, head alignment ≥ 0.900, hard-impact fraction ≤ 0.100, tool-acceleration p95 ≤ 120 m/s^2, cylinder headroom ≥ 0.005 m, and safety-termination rate ≤ 0.080. Final claims use six held-out meshes and report margins rather than pass/fail alone.

6.4 Clipped PPO objective

For observation o_t, the actor defines a Gaussian policy $\pi_\theta(a_t \mid o_t)$ and the critic estimates $V_\phi(o_t)$. With temporal-difference residual

$$\delta_t = r_t + \gamma V_\phi(o_{t+1}) - V_\phi(o_t), \tag{6}$$

generalized advantage estimation is

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l}. \tag{7}$$

The policy ratio and clipped surrogate are

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid o_t)}{\pi_{\theta_{\text{old}}}(a_t \mid o_t)}, \tag{8}$$

$$L^{\text{clip}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\Big)\Big]. \tag{9}$$

The optimized loss combines the negative clipped objective, clipped value loss and entropy regularization. The maintained configuration uses $\epsilon = 0.2$, $\gamma = 0.99$, $\lambda = 0.95$, five learning epochs, four minibatches, adaptive learning rate, desired KL 0.02 and gradient norm limit 1.0. Actor and critic are separate ELU MLPs with hidden widths 256–128–64 and normalized observations. Exact learning rate, entropy coefficient, initial standard deviation, environment count and iteration budget must be copied from the frozen experiment manifest, because these values changed during development.

6.5 Training protocol and the distillation finding

The strongest current policies were distilled from the DLS teacher. Repeated PPO fine-tuning produced a healthy critic but raised total reward by trading lateral tracking for additional contact;

it degraded the primary tracking objective. The thesis must not describe distillation merely as PPO initialization when the selected endpoint is the distilled actor.

For the final three-stack experiment, choose and predeclare one matched protocol:

- distill all three actors with identical teacher data and budgets, then evaluate; or
- distill identically and apply the same bounded PPO fine-tuning budget to all three, reporting both pre- and post-fine-tune checkpoints.

Do not compare a distilled baseline with a separately optimized adaptive policy and attribute the whole difference to the AAM. Report reward decomposition, critic explained variance, policy entropy/standard deviation and gate metrics during training so reward improvement cannot hide task regression.

7 Research questions and testable hypotheses

RQ1 – actuator representation. How do the frozen command ROM and frozen ID–FD pair differ in system response and closed-loop task performance under a matched policy interface?

RQ2 – identifiable context. Does the randomized deployment dataset contain a compact factor-dependent signal that improves held-out hydraulic rollout? Stage 22 already answers this positively for privileged z.

RQ3 – deployable inference. Can a causal AAM recover enough of the privileged latent advantage without pressure, physical factors, future state or hidden ROM state?

RQ4 – control adaptation. Does using $\hat{z}$ in a bounded ID adapter improve the frozen ID–FD stack on held-out hydraulic and mechanical configurations while preserving safety and nominal performance?

RQ5 – contribution location. How much of any improvement comes from the adapter, the inferred latent, direct policy conditioning or ordinary PPO compensation?

Primary hypotheses are:

- H_1: $SD_z(s, u, z_{\text{true}})$ outperforms matched SD_0 and latent controls on held-out extension and velocity rollouts (already supported at Stage 22);
- H_2: $SD_z(s, u, \hat{z})$ retains a statistically supported fraction of the true-latent gain on reserved runs;
- H_3: the true-latent ID adapter improves frozen ID–FD, establishing adapter capacity before AAM error is introduced;
- H_4: the AAM-latent adapter improves frozen ID–FD in paired held-out episodes and approaches the true-latent oracle;
- H_5: adaptation improves mismatch recovery without materially degrading the nominal domain or violating any of the eight gates.

8 Remaining implementation gates

Stage	Required work	Pass condition
23	Freeze the selected $d_z = 2$ encoder, SD_z, belief module, normalization, input mask, split and rollout protocol with hashes.	Frozen copies reproduce the Stage-22 metrics.
24	Build causal AAM histories from applied action, cylinder extension/velocity and joint position/velocity. Exclude factors, true latent, future state, hidden ROM state, differential pressure and task reference.	No leakage; windows remain within runs.
25	Compare predeclared histories such as 0.25, 0.5, 1.0 and 2.0 s using validation runs only.	Select the shortest history statistically comparable with the best downstream rollout.
26	Train $\hat{z} = AAM(H_t)$ from scratch. Measure latent error and stability, but select by SD_z rollout.	$\hat{z}$ beats zero/mean/shuffled controls downstream.
27	Compare SD_0, true-z SD_z and AAM-z SD_z on validation and reserved test runs.	Deployable $\hat{z}$ retains useful true-latent advantage.
28	Evaluate frozen historical ID transfer across nominal, hydraulic, mechanical, combined and held-out domains.	Transfer gap and failure modes are quantified before adapter training.
29	Train a small bounded ID residual adapter first with true z, then with $\hat{z}$. Keep ID and FD frozen.	True-z adapter helps; AAM version retains useful benefit.
30	Integrate causal AAM updates, history reset, warm-up, latency, action bounds and zero-adapter fallback in the ID–FD plant.	Stable online execution with no privileged deployment input.
31	Freeze all three policy stacks and matched manifests.	Only actuator/adaptation path differs; policy capacity and budget are matched.

9 Final comparative experiments

Every experiment uses paired factor configurations, terrain meshes, path families, seeds, episode lengths and safety stops for all three stacks. Report nominal, hydraulic-only, mechanical-only, combined and unseen-test domains separately.

9.1 Fixed comparison matrix

Every experiment in this section evaluates exactly these rows:

Stack	Frozen actuator components	Adaptive component
PPO–ROM	command ROM	none
PPO–ID–FD	control ID and strict FD	none
PPO–ID(AAM)–FD	control ID and strict FD	AAM plus bounded ID adapter

9.2 System response

Apply matched steps, ramps, PRBS and task commands. Per cylinder, report rise time, settling time, overshoot, steady-state error and gain; extension, velocity and pressure response; valve/action saturation; and cross-axis coupling. This experiment explains why one stack tracks laterally or vertically better rather than only reporting that it does.

9.3 Delay

Estimate command-to-pressure, command-to-velocity, pressure-to-velocity and, where meaningful, cylinder-to-tool response using controlled transients, cross-correlation and frequency/phase analysis. Report actuator, environment and AAM-update latencies separately. Do not label a static kinematic mapping as temporal delay.

9.4 Nonlinearity

Sweep command magnitude and sign, pressure, extension, velocity, payload, COM and arm posture. Quantify extension/retraction asymmetry, dead zone, saturation, pressure-dependent gain, load dependence, hysteresis and coupling using repeated paired measurements.

9.5 End-effector and mowing performance

Use the same feasible path families and held-out terrain meshes. Report lateral and cut-height RMSE, MAE and p95; maximum and along-track error; transient settling after path changes; contact fraction; head alignment; impacts; terminations; and results per family and worst mesh. The eight-gate contract is the minimum acceptance criterion, not a substitute for full distributions.

9.6 Robustness and generalization

Test in-range anchors and held-out combinations plus declared out-of-range stress tests. Add observation noise, action delay, sensor dropout, terrain shift, payload and COM changes. Separate interpolation from extrapolation. Report success, tracking degradation, saturation, safety terminations and adaptation recovery time relative to nominal.

9.7 Adaptation ablations

For **PPO–ID(AAM)–FD**, compare deployable $\hat{z}$, privileged true z, zero, mean, shuffled and random latents; frozen ID without adapter; adapter without AAM history; shorter/longer history; slower latent update; and optional policy conditioning on $\hat{z}$. The AAM claim requires deployable $\hat{z}$ to beat frozen ID–FD and approach the true-z oracle on held-out configurations.

9.8 Statistics

Predeclare primary metrics and seeds. Use paired episodes, per-run summaries and paired bootstrap confidence intervals. Keep validation choices separate from the untouched final test. Report failure cases, worst-case meshes and effect sizes, not only pooled means or binary gates.

10 Updated thesis chapter plan

Chapter	Purpose	Required content
1	Introduction	Hydraulic manipulation and verge-tracking problem; limitations of fixed learned actuators; research questions; contributions stated according to completed evidence rather than intention.
2	Fundamentals and related work	Hydraulic actuation, DLS, learned inverse/forward dynamics, reduced-order modeling, domain randomization versus adaptation, system identification, RMA-style context inference, residual control, PPO and safe evaluation.
3	System and problem formulation	FME 500/PhysX plant; four-cylinder coordinates; moving-verge primary task and parked diagnostic task; multi-rate timing; observable versus privileged signals; constraints; formal tracking and robustness objectives.
4	Data generation and preparation	Historical Simscape provenance for ROM/ID/FD; new IsaacLab 118-run collection; DLS and PRBS excitation; factor ranges; safety gates; 50 ms alignment; run-level splitting and design adequacy.
5	Learned models and privileged representation	Command ROM and ID/strict-FD contracts; SD_0; factor encoder and SD_z; Stage-22 controls and result; AAM dataset, history selection and downstream validation. Make clear that SD is not the deployed actuator.
6	Control framework and integration	The three descriptive policy stacks; bounded ID adapter; causal AAM update and fallback; 66-D observation; four-cylinder action mapping; bounded reward; clipped PPO objective; teacher distillation and matched training protocol.
7	Experimental evaluation and discussion	Model rollouts; ID transfer; system response; delay; nonlinearity; end-effector/mowing performance; robustness; adaptation ablations; statistics; mechanisms behind positive and negative findings.
8	Conclusion and outlook	Direct answers to each research question; demonstrated contributions; limitations of simulation, factor coverage, identifiability and policy optimization; real-system and uncertainty-aware extensions.

Recommended appendices contain coordinate/sign conventions and plant parameters; full dataset manifests and factor ranges; model architectures, normalizers and checkpoint hashes; complete PPO observations/actions/reward/hyperparameters; and per-seed/per-cylinder supplementary results.

11 Claim discipline and reproducibility checklist

Before final writing or a long experiment, verify all of the following:

- Use only **PPO–ROM**, **PPO–ID–FD** and **PPO–ID(AAM)–FD** as current method names and matrix rows.
- Do not state that the AAM conditions the ROM. In the primary design it conditions a bounded adapter around frozen ID; direct policy conditioning is an ablation.
- Do not describe the AAM or adaptive policy result as complete before Stages 23–31 pass and the closed-loop experiment is run.
- Keep command ROM, ID, strict FD, SD_0, SD_z, factor encoder and AAM contracts distinct. State which are frozen and which are trained from scratch.
- Never give physical factors, true latent, future state, hidden ROM state or hidden pressure to a deployable policy/AAM input.
- Preserve run-level splits and the reserved 16-run combined test set. Do not tune on it after Stage 22.
- Match policy architecture, task, terrain, action authority, observation content, reward, seeds and training budget across the three main stacks.
- Report the actual selected training method. Distillation and PPO fine-tuning are different procedures even when the resulting network has the same actor architecture.
- Report exact timing at each layer and verify action/target alignment at the model grid.
- Distinguish robust training across randomization from online adaptation to inferred context.
- Include negative results: ID–FD's contact failure, PPO fine-tuning's tracking trade, earlier latent redundancy and the need for wider matched excitation.

12 Current sources of truth

- Active policy comparison and repository overview: `README.md`, `PPO_ROM/`, `PPO_ID_FD/`.
- Deployment adaptation stages and final experiment matrix: `deployment_adaptation/README.md`.
- Stages 1–22 concise record: `AAM_module/memory.md` and individual stage documents.
- Frozen adaptation dataset: `AAM_module/datasets/master_v3_stage22_wide_frozen/`.
- Stage-22 pass: `outputs/sd_z_stage22_v6_wide/PASSED_GATE.json` and `outputs/sd_z_stage22_v6_wide/dz2_nodp_holdq/gate_verdict.json`.
- Runtime learned models: `rom_model/` and `id_fd_model/`.
- Mechanical/runtime implementation and timing: `mulag_plant/` and `verge_task/`.

Final thesis comparison: PPO–ROM versus **PPO–ID–FD** versus **PPO–ID(AAM)–FD**, evaluated under matched system-response, delay, nonlinearity, end-effector tracking, robustness and adaptation-ablation protocols.